UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2014/2015

APRIL EXAMINATION

UKFF4024 MULTINATIONAL FINANCE

FRIDAY, 8 MAY 2015 TIME: 2.30 PM - 5.30 PM(3.0 HOURS)

BACHELOR OF INTERNATIONAL BUSINESS (HONS)

Instructions to Candidates:

Section A: [Total: 40 marks]

This section consists of **ONE (1) compulsory** question that **MUST** be answered.

Section B: [Total: 60 marks]

This section consists of three (3) **optional** questions and **ONLY TWO (2)** questions are to be answered.

Note: For calculation question(s), marks will be awarded for detailed workings. For question(s) that require the use of financial calculator, marks will be awarded for steps showing input keyed in.

UKFF4024 MULTINATIONAL FINANCE

Section A **[Total: 40 marks]**

(Answer this ONE compulsory question)

Q1. a) Using the following table of swap rates, assume UTAR enters into a US$50 million seven-year cross-currency interest rate swap to pay euros and to receive dollars. The spot exchange rate at the time of the swap is US$1.16/€.

	Euro-€		*Swiss franc*		*U.S. dollar*		*Japanese yen*	
Years	*Bid*	*Ask*	*Bid*	*Ask*	*Bid*	*Ask*	*Bid*	*Ask*
1	2.99	3.02	1.43	1.47	5.24	5.26	0.23	0.26
2	3.08	3.12	1.68	1.76	5.43	5.46	0.36	0.39
3	3.24	3.28	1.93	2.01	5.56	5.59	0.56	0.59
4	3.44	3.48	2.15	2.23	5.65	5.68	0.82	0.85
5	3.63	3.67	2.35	2.43	5.73	5.76	1.09	1.12
6	3.83	3.87	2.54	2.62	5.80	5.83	1.33	1.36
7	4.01	4.05	2.73	2.81	5.86	5.89	1.55	1.58
8	4.18	4.22	2.91	2.99	5.92	5.95	1.75	1.78
9	4.32	4.36	3.08	3.16	5.96	5.99	1.90	1.93
10	4.42	4.46	3.22	3.30	6.01	6.04	2.04	2.07
12	4.58	4.62	3.45	3.55	6.10	6.13	2.28	2.32
15	4.78	4.82	3.71	3.81	6.20	6.23	2.51	2.56
20	5.00	5.04	3.96	4.06	6.29	6.32	2.71	2.76
25	5.13	5.17	4.07	4.17	6.29	6.32	2.77	2.82
30	5.19	5.23	4.16	4.26	6.28	6.31	2.82	2.88
LIBOR	3.0313	3.0938	1.3125	1.4375	4.9375	5.0625	0.1250	0.2188

Typical presentation by *The Financial Times*. Bid and ask spreads as of close of London business. US$ is quoted against three-month LIBOR; Japanese yen against six-month LIBOR; Euro and Swiss franc against six-month LIBOR.

i) Calculate all principal and interest payments in both currencies for the life of the swap. (14 marks)

ii) Assume that three years later, UTAR decides to unwind the swap agreement. If four-year fixed rates of interest in euros have now risen to 5.35% and four-year fixed rate dollars have fallen to 4.4%, and the current spot exchange rate is US$1.02/€, what is the net present value of the swap agreement? Who pays who and for what? (15 marks)

(b) UTAR and KTAR each seek funding at the lowest possible cost. UTAR would prefer the flexibility of floating rate borrowing, while KTAR wants the security of fixed rate borrowing. UTAR is the more creditworthy company. They face the following rate structure. UTAR, with the better credit rating, has lower borrowing costs in both types of borrowing:

	UTAR	KTAR
Credit rating	AA	BB
Fixed rate cost of borrowing	6%	10%
Floating rate cost of borrowing	KLIBOR + 0.5%	KLIBOR + 1.5%

UTAR wants floating rate debt, so it could borrow at KLIBOR + 0.5%. However, it could borrow fixed at 6% and swap for floating rate debt. KTAR wants fixed rate, so it could borrow fixed at 10%. However, it could borrow floating at KLIBOR + 1.5% and swap for fixed rate debt. What should UTAR and KTAR do? (11 marks)

[Total: 40 marks]

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Section B **[Total: 60 marks]**

(Answer any TWO (2) questions out of THREE (3) questions)

Q1. (a) Explain in detail the theory of International Fisher Effect. (12 marks)

(b) Assume that the Mexican peso has recently changed in value from MXP10.00/USD to MXP11.00/USD. Your home currency is the U.S. dollar. What is the percent change in the value of the Mexican peso in home currency terms versus in foreign currency terms? (8 marks)

(c) Differentiate between speculators and arbitrageurs in the FOREX market. (10 marks)

[Total: 30 marks]

Q2. (a) UTAR Corporation, a U.S car company acquired a Korean company that produces spare parts for automobile manufacturers. The purchase price was 7,030 million Won. 1,000 million Won has already been paid while the remaining 6,030 million Won is due in six months. Additional data is as follows:

Current spot rate (KRW/$)	1,200-1,210
6-month forward rate (KRW/$)	1,260-1,280
6-month Korean interest rate	16.00% p.a.
6-month U.S. interest rate	4.00% p.a.
6-month call option on Korean won at KRW1,200/$	3.0% premium
6-month put option Korean won at KRW1,200/$	2.4% premium

UTAR Corporation can invest at the rate given above, or borrow at 2% per annum above those rates. UTAR Corporation's weighted average cost of capital is 25%.

i) Calculate the payment in US$ if UTAR Corporation chooses not to hedge. (2 marks)

ii) Calculate the payment in US$ if UTAR Corporation uses forward marke the edge. (2 marks)

iii) Calculate the payment in US$ if UTAR Corporation uses money market hedge. (2 marks)

iv) Calculate the payment in US$ if UTAR Corporation uses option hedge. (5 marks)

v) Which is the best alternative? If the future spot rate moves in a favourable direction for UTAR Corporation, which hedging alternative still allow UTAR Corporation to benefit from it? Use break-even point calculation to illustrate. (5 marks)

Q2. (Continued)

(b) Provide a detailed explanation on the differences between transaction exposure, operating exposure, and translation exposure. (14 marks)

[Total: 30 marks]

Q3. (a) Trade can be induced by comparative advantage. Assume that China and France each have 1,000 production units. With one unit of production (a mix of land, labor, capital, and technology), China can produce either 10 containers of toys or 7 cases of wine. France can produce either 2 containers of toys or 7 cases of wine. Thus, a production unit in China is five times as efficient compared to France when producing toys, but equally efficient when producing wine. Assume at first that no trade takes place. China allocates 800 production units to building toys and 200 production units to producing wine. France allocates 200 production units to building toys and 800 production units to producing wine.

i) What is the production and consumption of China and France without trade? (6 marks)

ii) Assume complete specialization, where China produces only toys and France produces only wine. What would be the effect on total production? Explain. (9 marks)

(b) Floating rate loans are a widely used source of debt for firms worldwide. They are also the source of the single largest and most frequently observed corporate interest rate exposure. Assume that UTAR Corporation has taken out a US$10 million three-year floating rate loan which will be serviced with annual payments and total principal repayments at the end of the three-year period. The loan is priced at LIBOR + 1.500% and the LIBOR which is currently at 5.000% will be reset each year. When the loan is drawn down initially, at year zero, an up-front fee of 1.500% is charged by the lender. The fee results in a reduction in the net proceeds of the loan. Calculate the all-in-costs (AIC) of the loan under the following scenarios:

1) LIBOR remains the same,
2) LIBOR increases by 25 basis points per year, and
3) LIBOR decreases by 25 basis points per year.

(15 marks)

[Total: 30 marks]